

Explain the basic concepts of signals?

Event and signals:-

An Event is the specification of a significant occurrence that has a location in time and space.

In the context of state machines and an event is an occurrence of a stimulus that can trigger a state transition.

• A signal is a kind of event that represents significant of an asynchronous stimulus communicated b/w instances.

Kinds of Events:-

1) Events may be External or internal.

Event
Event declaration

`<<Signal>>`
off Hook

Idle.

off Hook / drop Connection()

Active.

Signal:-

Signal.

Parameters
`<<Signal>>`
Collision
force, F, level

`<<send>>`

Movement Agent
Position
Velocity
moveTo()

send dependency

Signal Event:-

A signal event represents a named object that is all dispatched (thrown) asynchronously by one object & then received (caught by) another. Exceptions are an example of internal signal. A signal event is an asynchronous event.

2) Signal Events may have instances, generalization relationships, attributes and operations.

3) A signal event may be sent as the action of a state transition in a state machine or the sending of a message in an interaction.

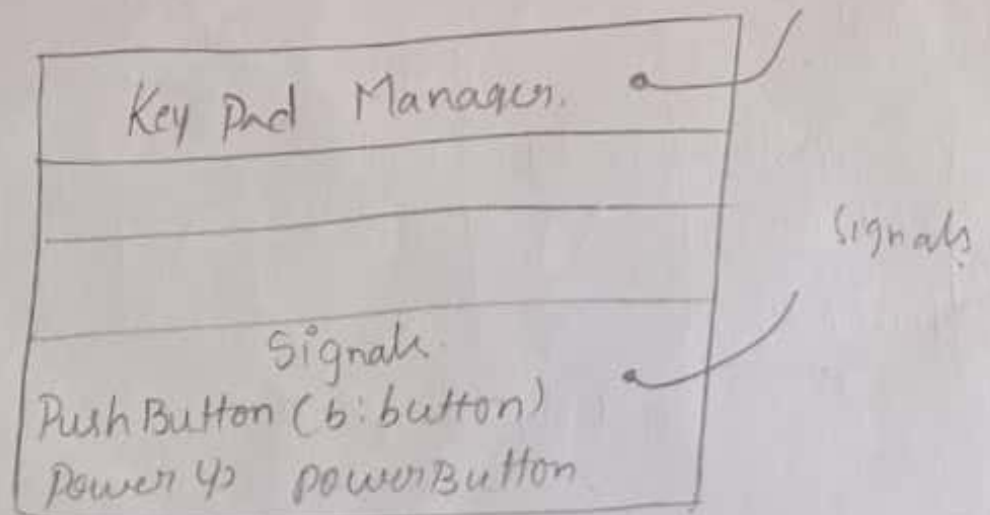
2) Explain basic concepts of the Events?

A. Event:- An event is the specification of a significant occurrence that has a location in time & space.

There are 3 types.

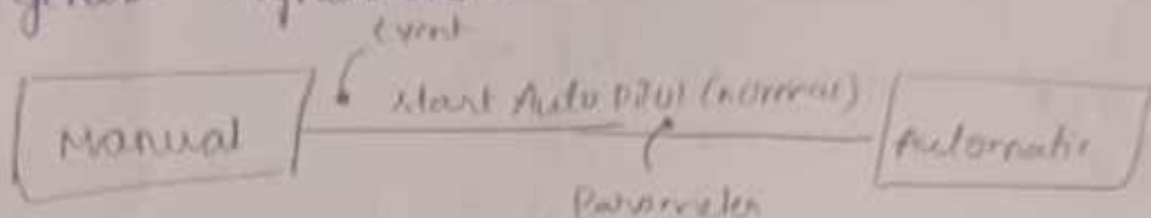
- 1) Signal Event.
- 2) Call Event
- 3) Time and change events.
- 4) Sending and Receiving Events.

Signals and Active classes.



all Events:-

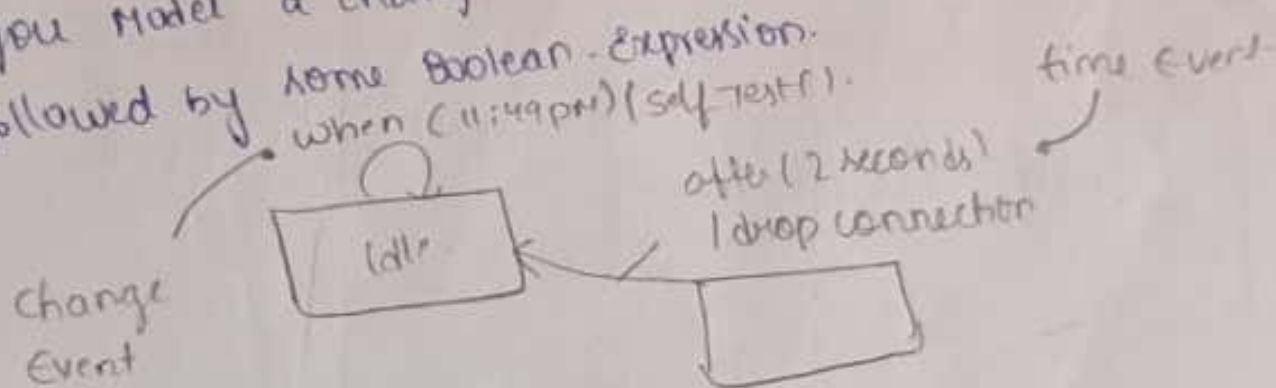
- 1) Just as a signal Event represents the occurrence of a signal a call Event represents the dispatch of an operation.
- 2) Whereas a signal is an asynchronous Event, a call Event is in general synchronous.



3) Time and change Events:

A time Event is an Event that represents the passage of time. In the UML you model a time Event by using the keyword after followed by some Expression that evaluates to a period of time.

A change Event is an Event that represents a change in state or the satisfaction of some condition. In the UML you model a change Event by using the keyword when followed by some Boolean Expression.



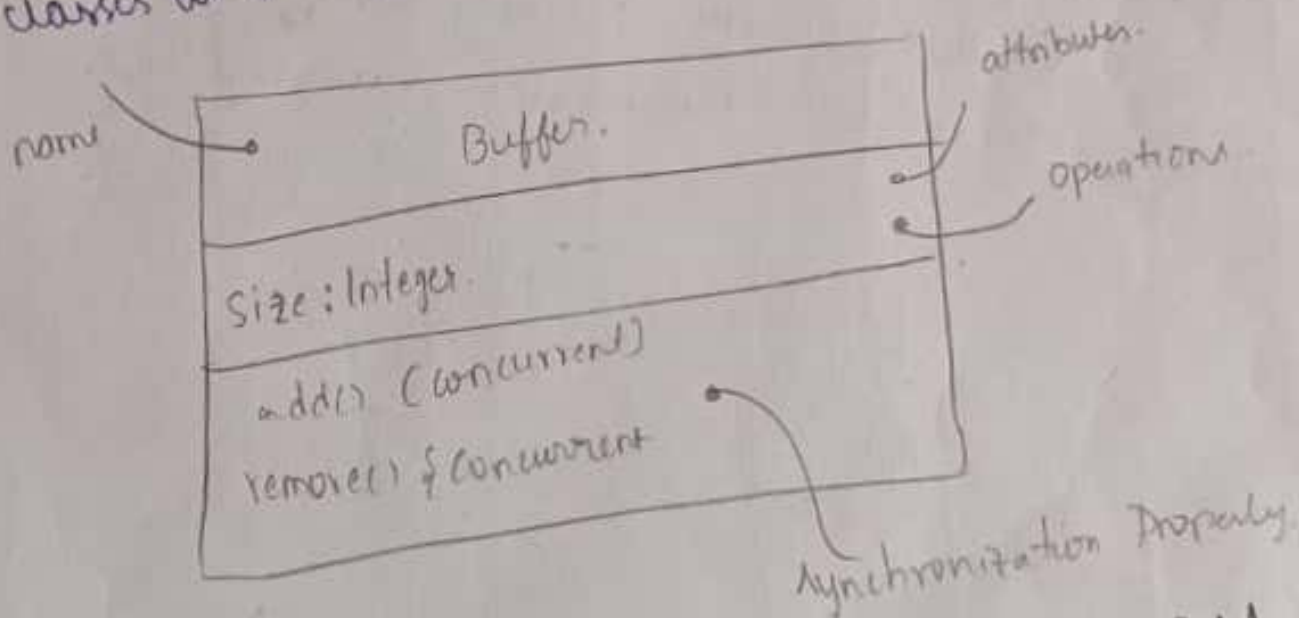
Explain the parts of a State?

States:-

A state is a condition or situation during the life of an object during which it satisfies some condition, performs some activity, or waits for some Event. An object remains in a state for a finite amount of

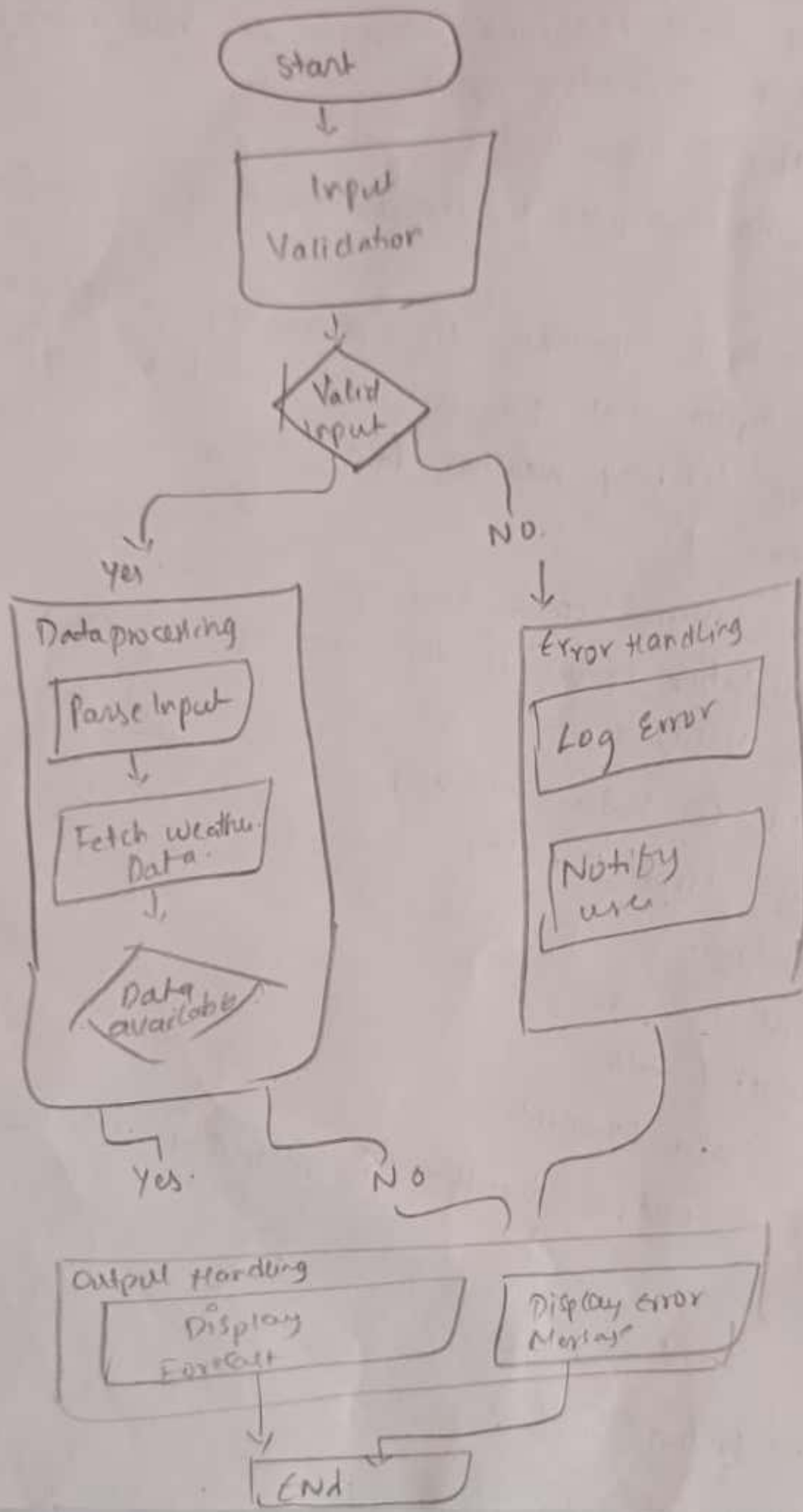
organize the 3 types of process synchronization in UML process synchronization:

- ① Synchronization means arranging the flow of objects so that mutual exclusion will be guaranteed.
- ② Three approaches are there to handle synchronization.
- ③ Sequential: callers must coordinate outside the object so that only one flow is in the object at a time.
- ④ Guarded - multiple flow of control is sequentialized with the help of object's guarded operations in effect it becomes sequential.
- ⑤ Concurrent: Multiple flow of control is guaranteed by treating each operation as atomic.
- ⑥ Synchronization are rendered in the operations of active classes with the help of constraints.



With a real diagram show how to model an API in a component diagram design.

The diagram depicts a high level system architecture where multiple clients (Mobile App, Web App, IoT devices)



6. Design the state machine diagram for Input Manager of a weather forecasting system.

A. Input Manager Flowchart:-

This flowchart represents the working of an Input Manager flow chart.

This flowchart represents the working of an Input Manager system that processes and validates user input for fetching weather data.

Flow process:

1. Start: The system begins execution.

2. Input Validation: Checks if the user input is valid.

3. Decision - Valid Input?

Yes: Proceeds to Data processing

No: Moves to Error Handling

4. Data processing:

Parser input.

Fetches weather data.

5. Decision - Data Available?

Yes: Moves to output handling (Displays the weather forecast).

No: Displays an error message.

6. End: the process concludes.